Data Analytics Project: Railway Gauge Trends

Dataset Used: Railway Gauge Data

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This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:
In this scenario, the railway gauge dataset is loaded into a Pandas DataFrame for analysis. The first five rows and column names are displayed to understand the structure of the dataset. Missing values are checked and replaced with zero. The Broad Gauge, Metre Gauge, Narrow Gauge, and Total columns are converted into numeric data types to prepare the dataset for further analysis.

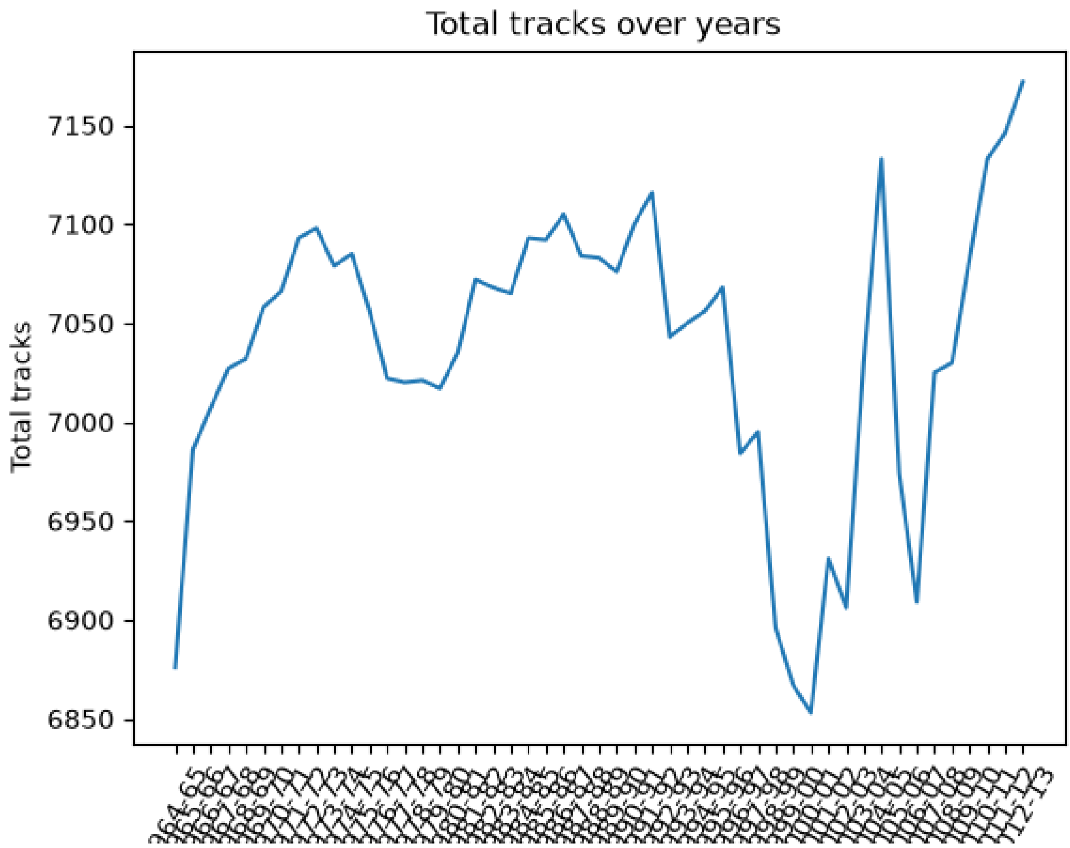
Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:
The dataset was successfully loaded and cleaned using Pandas. The structure of the dataset was examined by displaying the first five rows and column names. Missing values were identified and replaced with zero. The relevant gauge columns were converted into numeric types, making the dataset ready for further analysis.

Scenario 2: Simple Visualization (◍ Easy)

Description:
This scenario focuses on understanding the growth of the total railway tracks over the years. The Year and Total columns are extracted from the dataset and used to create a line graph. A suitable title and X-axis and Y-axis labels are added to make the visualization clear. The resulting graph is used to identify whether the overall railway track trend is increasing or decreasing.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.



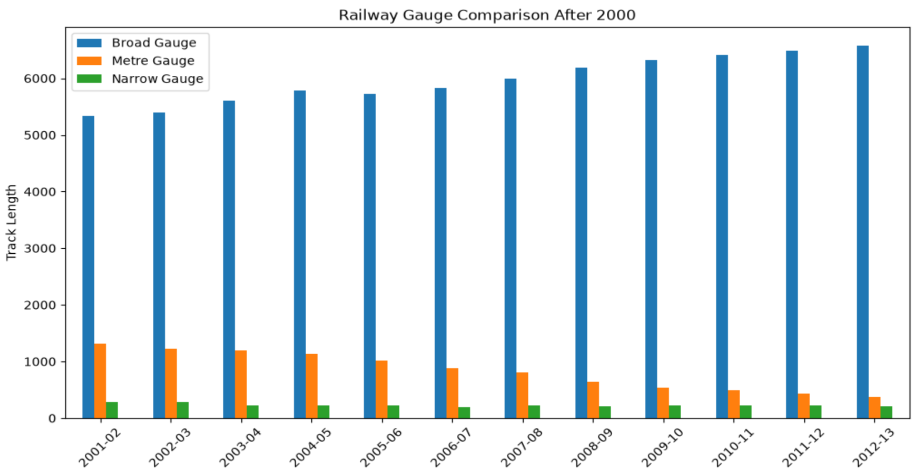
Conclusion:
The line graph provides a visual representation of the total railway tracks over the years. By observing the direction of the line, the overall trend can be identified as increasing or decreasing. This visualization provides a simple way to understand the historical growth of the railway network.

Scenario 3: Filtering + Bar Chart (▨ Medium)

Description:
This scenario focuses on analyzing railway gauge expansion in recent years. The dataset is filtered to include only years after 2000. Broad Gauge, Metre Gauge, and Narrow Gauge are selected from the filtered dataset. A grouped bar chart is created to compare the three gauge types. A legend and appropriate labels are added to make the comparison clear.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph



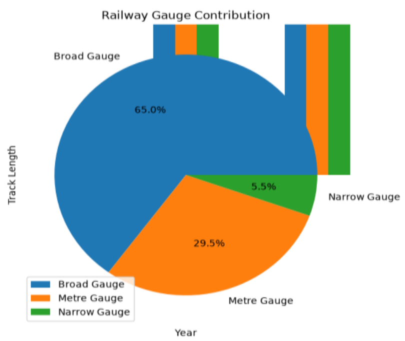
Conclusion:
The grouped bar chart provides a comparison of Broad Gauge, Metre Gauge, and Narrow Gauge after 2000. The visualization helps identify the gauge with the highest track length during the recent period. The gauge with the largest values can be considered the dominant gauge in the selected years.

Scenario 4: Feature Engineering + Pie Chart (🟠 Medium)

Description:
This scenario analyzes the contribution of different railway gauge types across all available years. The total sum of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated. These values are stored in a Pandas Series or DataFrame. A pie chart is then created to show the percentage contribution of each gauge type, with percentage labels added using `autopct`.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph



Conclusion:
The pie chart provides a visual comparison of the contribution of the three railway gauge types across all years. The gauge occupying the largest portion of the pie chart contributes the most to the analyzed railway gauge data. This helps identify the gauge with the highest overall contribution.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

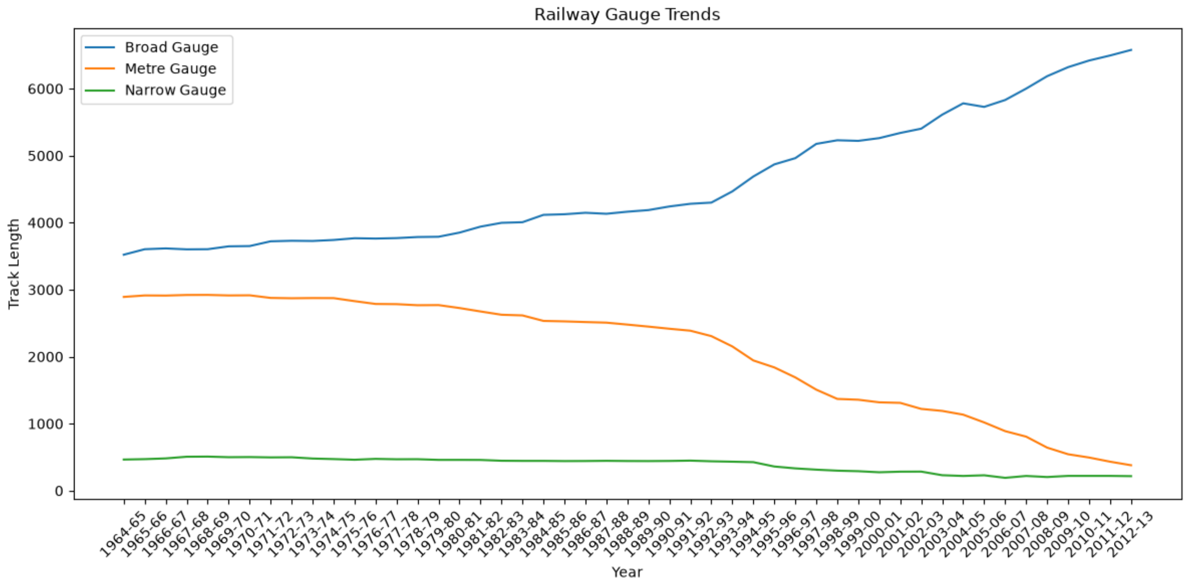
Description:
This scenario performs an advanced analysis of railway gauge trends. New percentage columns are created to represent the composition of Broad Gauge, Metre Gauge, and Narrow Gauge relative to the Total tracks. NumPy's `np.diff()` function is used to calculate yearly growth in Total tracks. Line and stacked bar charts are created to analyze gauge trends and composition. The analysis also identifies years with the highest growth and periods where a gauge experienced decline.

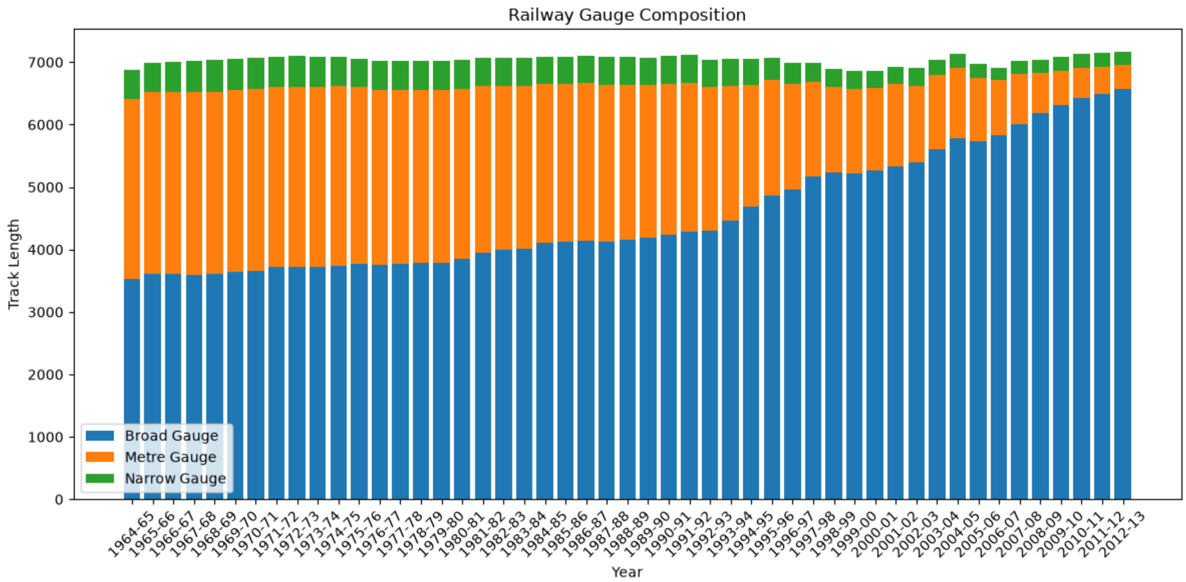
Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graph

Line Graph: Shows the growth and decline trends of Broad, Metre, and Narrow Gauge over the years.

Stacked Bar Chart: Shows the composition and contribution of the three gauge types across different years.





Conclusion:

The advanced analysis provides a detailed view of railway gauge trends and yearly changes. Percentage composition helps compare the relative contribution of each gauge, while yearly growth identifies periods of expansion or decline. The results can be used to determine whether the railway system is shifting toward a single dominant gauge based on the observed gauge composition trends.

Final Conclusion:

This project analyzes railway gauge data through a series of data processing, numerical analysis, and visualization tasks. The project covers data loading, missing value handling, filtering, aggregation, feature engineering, yearly growth calculation, and multiple types of graphs. The analysis helps understand railway track growth and the contribution of different gauge types over time. The final analysis can be used to determine whether the railway system is shifting toward a single dominant gauge.

What you have learned

Through this project, I learned how to use Pandas for data loading, cleaning, filtering, and aggregation. I learned how to use NumPy for numerical calculations such as yearly growth using `np.diff()`. I also learned how to create line graphs, grouped bar charts, pie charts, and stacked bar charts using Matplotlib. This project helped me understand how data processing and visualization can be combined to analyze trends and draw conclusions from a dataset.